

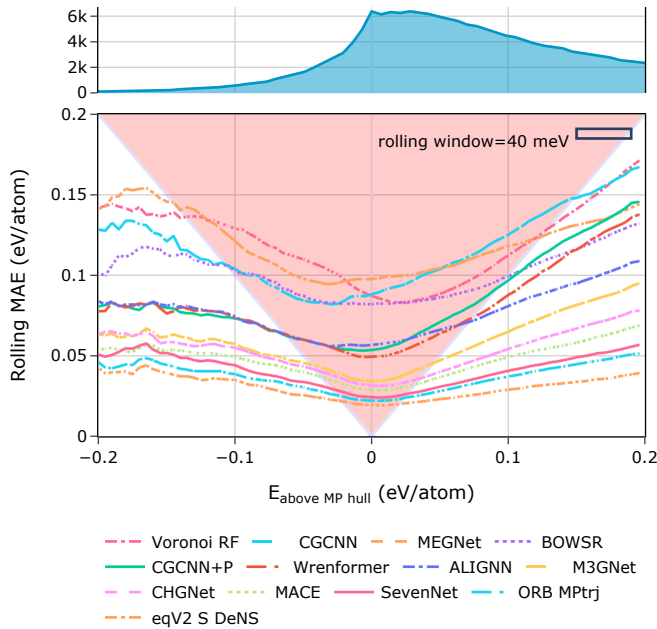


Figure 3: Universal potentials are more reliable classifiers because they exit the red triangle earliest. These lines show the rolling MAE on the WBM test set as the energy to the convex hull of the MP training set is varied. Lower is better. Inside the large red ‘triangle of peril’, models are most likely to misclassify structures. As long as a model’s rolling MAE remains inside the triangle, its mean error is larger than the distance to the convex hull. If the model’s error for a given prediction happens to point towards the stability threshold at $E_{\text{above MP hull}} = 0$, its average error will change the stability classification from true positive or negative to false negative or positive. The width of the ‘rolling window’ box indicates the width over which prediction errors were averaged.

Figure 3 provides a visual representation of the reliability of different models as a function of a material’s DFT distance to the MP convex hull. The lines show the rolling mean absolute error (RMAE) of model-predicted hull distances vs DFT. The red-shaded area, which we coin the ‘triangle of peril’, emphasizes the zone where the average model error surpasses the distance to the stability threshold at 0 eV. As long as the rolling MAE remains within this triangle, the model is highly susceptible to misclassifying structures. The average error in this region is larger than the distance to the classification threshold at 0, consequently in cases where the error points towards the stability threshold it would be large enough to flip a correct classification into an incorrect one. The sooner a model exits the triangle on the left side (negative DFT hull distance), the less likely it is to incorrectly predict stable structures as unstable, thereby reducing false negatives. Exciting early on the right side (positive DFT hull distance) results in a lower likelihood of predicting unstable

structures as stable, leading to fewer false positives.

All models tend to have lower rolling error towards the plot’s left edge compared to the right edge. This imbalance shows that models are more prone to false positive than false negative predictions. In other words, all models are less likely to predict a material at -0.2 eV/atom DFT hull distance as unstable than a material at 0.2 eV/atom DFT hull distance as stable. From a practitioner’s standpoint, this is undesirable as attempting to validate an unstable material usually has a much higher opportunity cost than missing a stable one. We hypothesize this error imbalance is due to the MP training set having an uncharacteristically high share of stable materials, causing statistical models trained on it to be biased towards making low-energy predictions even for high-energy atomic configurations. Training on datasets with more high-energy structures, such as Alexandria [7] and OMat24 [57], would be expected to improve performance by balancing out this source of bias.

4 Discussion

We have demonstrated the effectiveness of ML-based triage in high-throughput materials discovery and posit that the benefits of including ML in discovery workflows now clearly outweigh the costs. Table 1 shows that in a realistic benchmark scenario that several models achieve a discovery acceleration greater than 2.5 across the whole dataset and up to 6 when considering only the 10k most stable predictions from each model (table S2). When starting this project, we were unsure which is the most promising ML methodology for high-throughput discovery. Our findings demonstrate a clear superiority in the accuracy and extrapolation performance of UIPs. Modeling forces enables these models to chart a path through atomic configuration space closer to the DFT-relaxed structure from where a more informed final energy prediction is possible.

As the convex hull becomes more comprehensively sampled through future discoveries, the fraction of unknown stable structures will naturally decline. This will lead to less enriched test sets and, consequently, more challenging and discriminative discovery benchmarks. However, the discovery task framed here addresses only a limited subset of potential UIP applications. We believe that additional benchmarks are essential to effectively guide UIP development. These efforts should prioritize task-based evaluation frameworks that address the four critical challenges we identify for narrowing the deployment gap: adopting prospective rather than retrospective benchmarking, tackling relevant targets, using informative metrics, and scalability.

Looking ahead, the consistently linear log-log learning curves observed in related literature [74] suggest that further decreases in the error of UIPs can be readily unlocked with increased training data. This has been borne out in the

scaling results of GNoME [62], MatterSim [75], Alexandria [7], and OMat24 [57] which all show improvements in performance when training on much larger datasets. To realize the full potential of scaling these models, future efforts should deploy their resources to generate large quantities of higher-than-PBE fidelity training data. The quality of a UIP model is circumscribed by the quality and level of theory of its training data.

Beyond simply predicting thermodynamic stability at zero Kelvin, future models will need to understand and predict material properties under varying environmental conditions, such as finite temperature and pressure, to aid in materials discovery. In this context, temperature-dependent dynamical properties constitute an area ripe for interatomic potentials. Another key open question is how effectively these models can contribute to the computational prediction of synthesis pathways. Many existing approaches for reaction pathway prediction involve the use of heuristic rules for dealing with the significant added complexity of metastability alongside traditional ground state ab-initio data [76–78]. These algorithms will massively benefit from more efficient estimates of reaction energy barriers [79] and non-crystalline, out-of-equilibrium materials [80], opening up a whole new field to machine learning accelerated inquiry.

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6 Author Contributions

Janosh Riebesell: Methodology, Software, Data Curation, Investigation (Training: CGCNN, CGCCN+P, Wrenformer, Voronoi RF), Validation, Formal Analysis, Writing – original draft. Rhys Goodall: Conceptualization, Software, Validation, Formal Analysis, Writing – original draft, review & editing. Philipp Benner: Software, Investigation (Training: ALIGNN, MACE), Writing – original draft. Yuan Chiang: Investigation (Training: MACE), Formal Analysis, Writing – review & editing. Bowen Deng: Data Curation (MPtrj), Investigation (Training: CHGNet), Writing – review & editing. Gerbrand Ceder: Supervision, Funding Acquisition. Mark Asta: Supervision, Funding Acquisition. Alpha Lee: Supervision. Anubhav Jain: Supervision. Kristin Persson: Supervision, Writing – review & editing, Funding Acquisition.

7 Code availability

We welcome further model submissions to our GitHub repository <https://github.com/janosh/matbench-discovery>.

8 Data availability

We chose the latest Materials Project (MP) [3] database release (v2022.10.28 at time of writing) as the training set and the WBM dataset [40] available at <https://figshare.com/articles/dataset/22715158> as the test set for this benchmark. A snapshot of every ionic step including energies, forces, stresses and magnetic moments in the MP database is available at <https://figshare.com/articles/dataset/23713842>.

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